PRACTICAL 9-

Aim- A system has RAM which can store four pages simultaneously to satisfy page requests. Write a Program to calculate percentage page hit and page fault if the system uses Optimal page replacement technique for the page references entered by the user i.e. 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 (No of Frames=3).

Code-

```
#include<stdio.h>

int main()
{
    int pages[] = {1,2,3,4,1,2,5,1,2,3,4,5};
    int frames[3];
    int i,j,k,n=12,framesize=3;
    int hit=0,fault=0,found,pos;

    for(i=0;i<framesize;i++)
        frames[i] = -1;

    for(i=0;i<n;i++)
    {
        found = 0;

        for(j=0;j<framesize;j++)
        {
            if(frames[j] == pages[i])
            {
                hit++;
                found = 1;
                break;
            }
        }

        if(found == 0)
        {
            int farthest = i, index = -1;

            for(j=0;j<framesize;j++)
            {
                for(k=i+1;k<n;k++)
                {
                    if(frames[j] == pages[k])
                    {
                        if(k > farthest)
                        {
                            farthest = k;
                            index = j;
                        }
                        break;
                    }
                }
                if(k == n)
                {
                    index = j;
                    break;
                }
            }

            if(index == -1)
                index = 0;

            frames[index] = pages[i];
            fault++;
        }
    }

    printf("Total Page Hits = %d\n", hit);
    printf("Total Page Faults = %d\n", fault);

    printf("Hit Percentage = %.2f%%\n", (hit/(float)n)*100);
    printf("Fault Percentage = %.2f%%\n", (fault/(float)n)*100);

    return 0;
}

^G Help          ^O Write Out     ^F Where Is      ^K Cut           ^T Execute
```

Output-

```
HP@DESKTOP-B4BD73N MINGW64 ~
$ nano optimal.c

HP@DESKTOP-B4BD73N MINGW64 ~
$ gcc optimal.c -o optimal

HP@DESKTOP-B4BD73N MINGW64 ~
$ ./optimal
Total Page Hits = 5
Total Page Faults = 7
Hit Percentage = 41.67%
Fault Percentage = 58.33%
```